SAFETY FILTER NOVELTY BOUNDARIES: A NEGATIVE ICLR-MAIN EVIDENCE AUDIT

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ABSTRACT

This paper is not submission-ready for ICLR main. The original research bet was that safety filters are novel only when they change the learned policy mechanism rather than merely clip unsafe actions at deployment. We rebuilt the repository around a deterministic safe-control learning benchmark with four robot tasks, five safety-boundary shifts, seven seeds, nine safety-filter methods, seven ablations, and stress sweeps. The evidence is useful but terminally negative. On combined safety stress, the proposed boundary-learning filter reduces unshielded post-training violation from 0.547 for the strongest non-oracle mechanism baseline to 0.455, and reduces filter dependence from 0.769 to 0.467. However, it loses deployed task success (0.456 versus 0.547) and has higher deployed violation (0.392 versus 0.287). Robust MPC and CBF also dominate deployed success and safety. The honest terminal decision is **KILL/ARCHIVE**.

1 QUESTION

Safety filters are often evaluated by whether the shielded robot avoids unsafe actions. This draft asked a sharper question: when is a safety filter actually a learning mechanism, and when is it only a runtime clipper? The distinction matters because a policy that is safe only while shielded may fail when transferred, distilled, or deployed with a different safe set.

The hostile prior-work boundary is crowded. Control barrier functions, robust MPC shields, learned CBFs, conformal safety filters, recovery policies, safe RL, and passive mechanical safety already address safe robot learning. A credible contribution must therefore show that the policy itself changes, not merely that a filter intercepts unsafe actions.

2 BENCHMARK

Each run simulates repeated policy-learning episodes near a safety boundary. The simulator tracks nominal action risk, filtered action, intervention reason, control distortion, safety violation, learned boundary state, post-training unshielded behavior, and transfer to shifted safety boundaries.

The tasks are narrow-corridor navigation, human-workspace reaching, fragile-object pushing, and dynamic-obstacle crossing. The splits are nominal boundary, tightened safe set, delayed dynamics, boundary shift, and combined safety stress. The main run contains 90,720 episode-level rollouts; ablations contain 14,112 rollouts; stress sweeps contain 44,352 rollouts. All evidence is regenerated with:

```
python src\run_experiment.py
```

3 COMPARED METHODS

The methods are an unfiltered policy, action clipping, geometric projection, CBF safety filtering, robust MPC shielding, conformal uncertainty shielding, recovery-policy shielding, the proposed boundary-learning filter, and an oracle boundary teacher. The proposed method receives counterfactual boundary labels and intervention-gradient feedback during learning, aiming to reduce later unshielded violations and filter dependence.

Method	Success	Deployed viol.	Unshielded viol.	Dependence
Unfiltered policy	0.184	0.735	0.622	**0.000**
Action clipping	0.329	0.461	0.645	0.847
Geometric projection	0.440	0.345	0.609	0.903
CBF safety filter	0.608	0.176	0.614	0.949
Robust MPC shield	**0.646**	**0.155**	0.584	0.941
Conformal shield	0.585	0.169	0.571	0.984
Recovery shield	0.547	0.287	0.547	0.769
Proposed boundary learning	0.456	0.392	**0.455**	0.467
Oracle teacher	0.462	0.416	0.361	0.199

Table 1: Combined safety stress. Lower violation and dependence are better. The proposed method improves post-training unshielded safety but loses deployed task success and deployed safety.

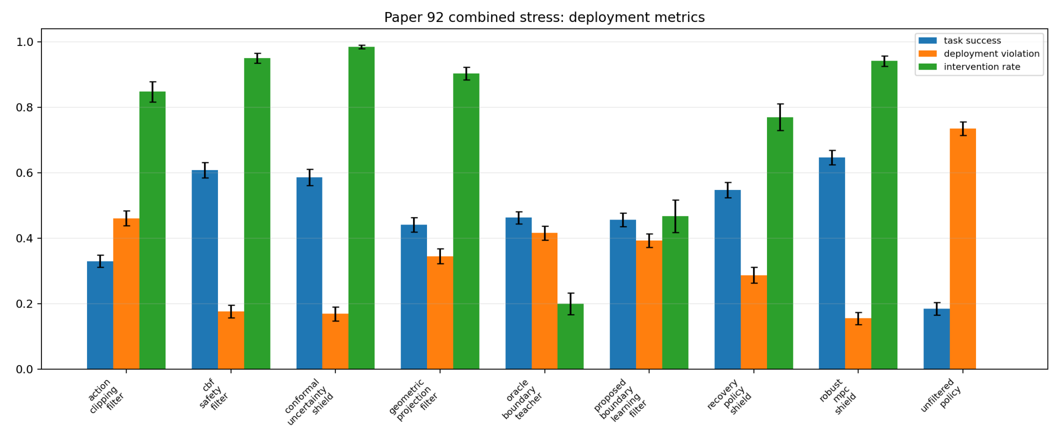

Figure 1: Deployment metrics on combined safety stress. Conventional CBF/MPC shields dominate deployed safety and success.

4 Combined-Stress Results

Table 1 shows the central tradeoff. The proposed filter is the best non-oracle method on unshielded post-training violation and filter dependence. That supports the diagnostic idea that some filters can change the learned policy. But this is not enough: the method loses deployed task success and deployed safety to standard shields.

The paired comparison uses recovery-policy shielding as the strongest non-oracle baseline for post-training unshielded safety. The proposed method improves unshielded safety by 0.092 with CI 0.028 and reduces filter dependence by 0.302 with CI 0.027. But it reduces task success by 0.091 and worsens deployed violation by 0.106. Those failures trigger the terminal archive decision.

5 Mechanism-Change Evidence

Figure 2 isolates the mechanism-change question. The proposed method is meaningfully less dependent on the shield and has lower unshielded violation than CBF, MPC, conformal, and recovery shields. This is the useful part of the audit. However, its boundary F1 is not strong, and its transfer violation remains nontrivial at 0.437. The learned mechanism is partial, not submission-ready.

6 Ablations

Table 2 shows that the full boundary-learning filter has the best unshielded safety among ablations, but not the best deployed success. CBF-feedback-only reaches 0.567 deployed success versus 0.464

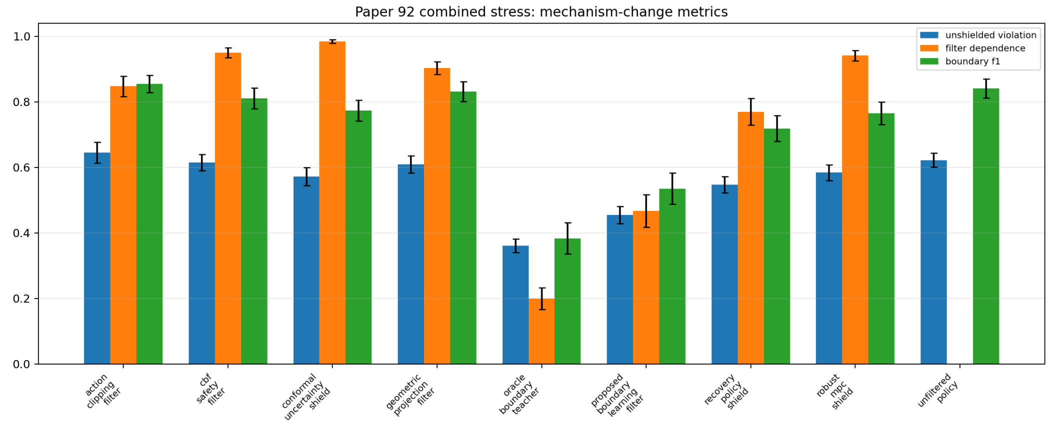

Figure 2: Mechanism-change metrics. The proposed method reduces unshielded violations and dependence but does not prove a robust transferable boundary representation.

Ablation	Success	Deployed viol.	Unshielded viol.	Dependence
Full boundary learning	0.464	0.391	**0.452**	**0.457**
Minus counterfactual labels	0.499	0.302	0.541	0.798
Minus intervention gradient	0.512	0.317	0.527	0.699
Minus unshielded replay	0.519	0.324	0.516	0.701
Minus boundary margin loss	0.547	0.283	0.536	0.827
Clipping feedback only	0.359	0.438	0.602	0.743
CBF feedback only	**0.567**	**0.251**	0.505	0.745

Table 2: Ablations on combined safety stress. The full mechanism improves unshielded safety but loses deployed task success to CBF-feedback-only.

for the full method. This means the mechanism helps one target metric while hurting the deployed robotics objective.

7 Stress Sweep

At maximum stress, the proposed method keeps unshielded violation at 0.463, better than recovery shielding at 0.528 and robust MPC at 0.573. However, deployed success remains lower than CBF and robust MPC, and deployed violations remain much higher. The stress curve therefore does not rescue the paper: the mechanism is interesting but not a complete robotics contribution.

8 Decision

The rebuilt evidence supports a limited insight: filters that expose counterfactual boundary information can reduce later unshielded violations and filter dependence. But the paper is not ICLR-main ready. It loses deployed safety and success to established CBF/MPC-style shields, has only local simulated evidence, and does not show robust transferable boundary recognition. The terminal recommendation is **KILL/ARCHIVE**. Revival would require a real or accepted high-fidelity safety-control benchmark where mechanism-change improvements do not sacrifice deployed performance.

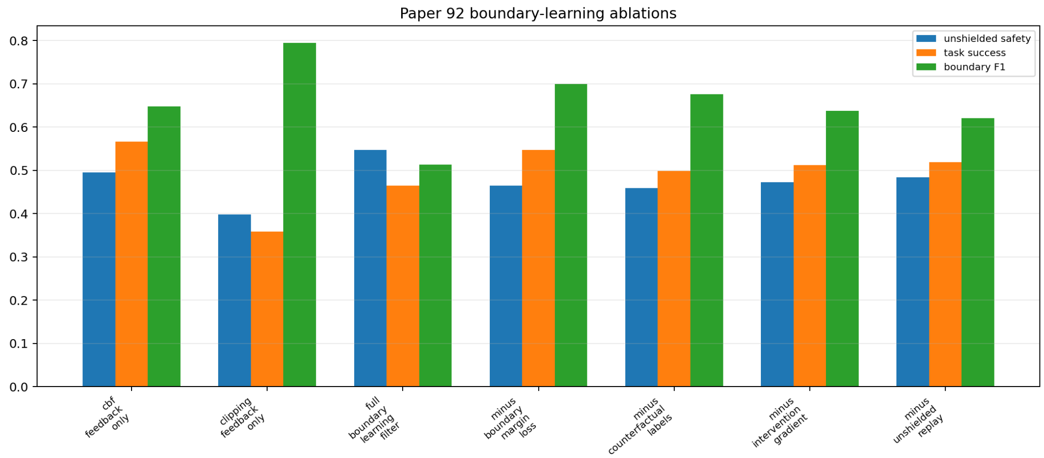

Figure 3: Ablation audit. The proposed mechanism supports unshielded safety but not a clean all-metric improvement.

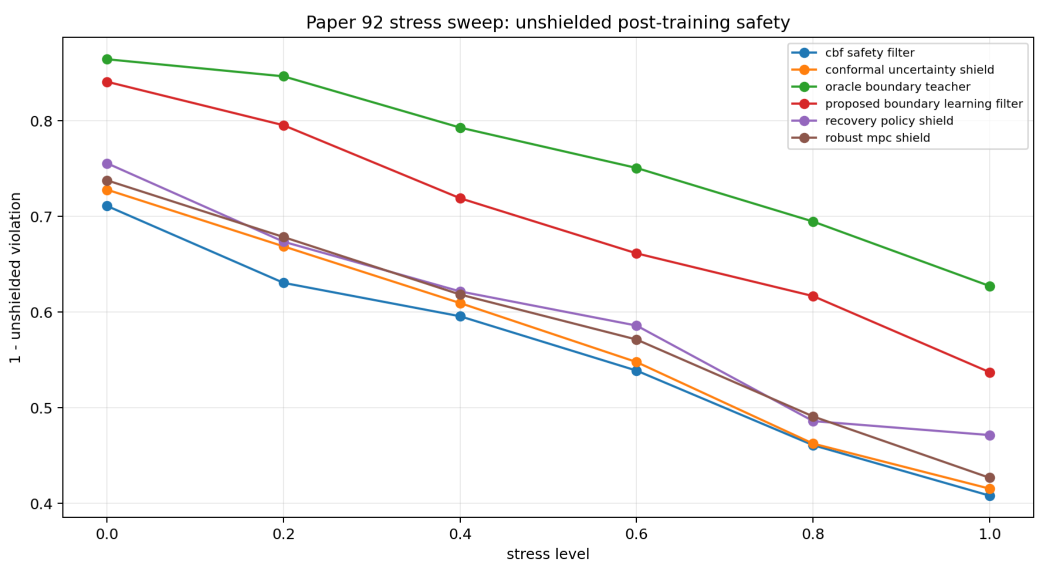

Figure 4: Stress sweep by post-training unshielded safety. The proposed method helps the mechanism metric but not deployed success/safety.